Joseph Sackitey

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Education

Gettysburg College — Gettysburg, PA
B.S. Computer Science and Physics — *Expected May 2027*

- **Relevant Coursework:** Algorithms, Linear Algebra, Data Structures, Operating Systems, Machine Learning, Databases (PostgreSQL, MySQL), Computer Organization & Assembly, Software Engineering
- **GPA:** 3.81

Technical Skills

Languages: Java, Python, C/C++, TypeScript, C#, Kotlin, SQL, Assembly Language
Data & Backend: Apache Flink, Kafka, Avro, PostgreSQL, Redis, GraphQL, REST APIs
Cloud & Developer Tools: Linux/Unix, AWS (ECS, RDS, ElastiCache), GCP, Docker, Kubernetes, Jenkins, Maven, JUnit, Git, CI/CD, TDD, Agile/Scrum

Professional Experience

Intuit — May 2026 – Aug 2026
Data Engineering Intern — *Mountain View, CA*

- Designed and deployed to production a multi-agent AI plugin for Claude Code and Cursor across 5 Data Engineering teams, automating Apache Flink pipeline development from Jira ticket to release-ready PR and cutting dev time from 2 days to 1 hour (48x faster).
- Built and deployed to production a CI/CD deployment and validation framework for processor artifacts across four critical marketing data pipelines, tripling developer productivity and enabling reliable, self-serve releases.
- Built an end-to-end CI/CD test framework for the TurboTax filing journey, validating enriched events across 4 pipelines and enforcing automated release gates in QAL, E2E, and production on every pull request and master merge, reducing validation turnaround by 83% (3 hours to 30 minutes).

Google — May 2025 – Aug 2025
Associate Software Developer Intern — *San Francisco, CA*

- Built a new high-performance Windows desktop app for Google Play Games on PC end-to-end, reducing launch latency by 10% and decreasing installer payload by 70 MB compared to the legacy client baseline.
- Enhanced web infrastructure and cross-process communication between the desktop client and the Progressive Web App, reducing memory use, CPU load, and launch time.
- Implemented a WebSocket-based IPC layer in C# and TypeScript, enabling seamless real-time state synchronization between native desktop components and the PWA frontend.

Gettysburg College — Aug 2026 – Present
Computational Physics Researcher — *Gettysburg, PA*

- Engineered an end-to-end Python/C++ fluid dynamics simulation pipeline modeling hydrodynamic propulsion and collective motion for $O(10^4)$ interacting particles.
- Developed an automated 3D geometric packing generator enforcing periodic boundary conditions, spatial clearance validation, and collision/overlap detection.
- Architected data analytics and visualization workflows to calculate trajectory diffusion metrics (MSD) and render interactive HTML/MP4 simulations.

Projects

RxConnect (TypeScript, Next.js, PostgreSQL, Redis, Docker, AWS)

- Architected a multi-tenant digital health platform connecting doctors, pharmacies, and patients, supporting real-time prescription routing and role-based access across 3 portals using an Nx monorepo, Next.js 14, and NestJS microservices.
- Built a scalable backend with PostgreSQL + PostGIS for geospatial pharmacy search and BullMQ/Redis for asynchronous SMS notifications, deployed on AWS (ECS Fargate, RDS, ElastiCache) with automated CI/CD.

ML Code Review Buddy (TypeScript, Flask, LangChain, NVIDIA Llama Nemotron)

- Co-developed AI-powered VS Code extension that analyzes Jupyter notebooks and delivers real-time ML optimization suggestions, reducing code review time by approximately 40%.
- Architected TypeScript frontend and Flask backend with LangChain-powered NVIDIA Llama integration for context-aware code recommendations.

Leadership, Honors & Awards

Generation Google Scholar (IIE) – $10,000 award recognizing leadership excellence and commitment to diversity in technology.

John Alfred Hamme Award (Gettysburg College) – Honored for exemplary student leadership, citizenship, and community impact.

2x Hackathon Winner (University of Delaware); ColorStack, ACM, CodePath member; Resident Assistant.